Wave Properties

PHYS 202: Introductory Physics II

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Purpose. These notes introduce the main wave concepts and algebra for Week 2: wave classification, wave speed on a string, interference, cable applications, standing waves, and harmonics.

Topics at a glance

Topic	Central question	Key tool
Wave type	How does the medium move relative to the wave?	Perpendicular or parallel
String-wave speed	How do tension and linear density affect speed?	$\mu = m/L$ and $v = \sqrt{F_T/\mu}$
Interference	What happens when waves overlap?	Superposition and relative phase
Cable applications	How can a traveling pulse reveal the tension?	$v = L/\Delta t$ and $F_T = \mu v^2$
Standing waves	Which resonant patterns fit on a string?	$f_n = nf_1$

Basic wave language

Quantity	Meaning	SI unit
Amplitude A	Maximum displacement from equilibrium	m
Wavelength λ	Length of one complete spatial cycle	m
Period T_p	Time for one complete oscillation	s
Frequency f	Number of complete oscillations per second	Hz
Wave speed v	Speed at which the disturbance travels	m/s

The basic relationships are

$$\boxed{f = \frac{1}{T_\text{p}}} \qquad \text{and} \qquad \boxed{v = \lambda f}. \tag{1}$$

1 Identifying the Type of Wave

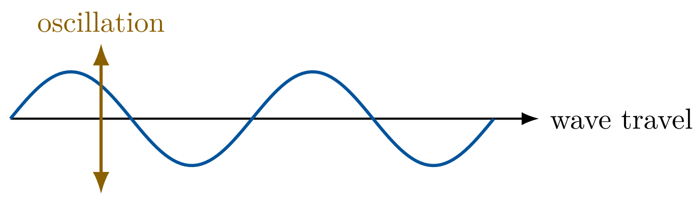
A wave transfers a disturbance and energy from one place to another. In a mechanical wave, each part of the medium moves locally around its equilibrium position; the material does not travel along with the wave over a large distance.

To classify a wave, compare two directions:

1. the direction in which the disturbance travels;
2. the direction in which the medium oscillates.

1.1 Transverse waves

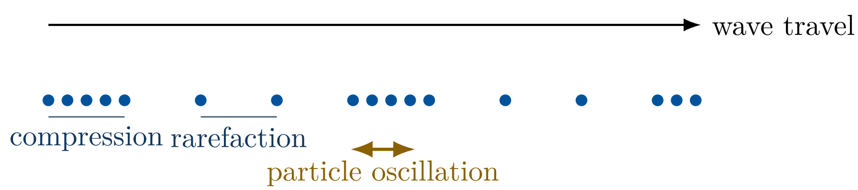
In a **transverse wave**, the medium oscillates **perpendicular** to the direction of wave travel.



A pulse on a stretched string is transverse: the pulse may travel horizontally while pieces of the string move vertically. Light is also transverse, although light is an electromagnetic wave and does not require a material medium.

1.2 Longitudinal waves

In a **longitudinal wave**, the medium oscillates **parallel** to the direction of wave travel.



Sound in air is longitudinal. Air molecules move back and forth, producing alternating crowded regions called **compressions** and spread-out regions called **rarefactions**.

2 Wave Speed on a String

The speed of a transverse wave on a stretched string depends on the string's tension and its mass per unit length.

The **linear mass density** is

$$\boxed{\mu = \frac{m}{L}}, \tag{2}$$

where m is the mass of a length L of string. Its SI unit is kg/m. The wave speed is

$$\boxed{v = \sqrt{\frac{F_T}{\mu}}}, \tag{3}$$

where F_T is the string tension. Substituting $\mu = m/L$ gives

$$\boxed{v = \sqrt{\frac{F_T L}{m}}}. \tag{4}$$

Physically,

- greater tension makes the disturbance travel faster;
- greater mass per unit length makes the disturbance travel slower;
- the square root reduces the size of either effect.

2.1 A reliable scaling method

Suppose a change multiplies the tension, mass, and length by factors a, b, and c:

$$F_T' = aF_T, \qquad m' = bm, \qquad L' = cL.$$

First update the linear density:

$$\mu' = \frac{m'}{L'} = \frac{b}{c}\mu.$$

Then compare the speeds:

$$\boxed{\frac{v'}{v} = \sqrt{\frac{ac}{b}}}. \tag{5}$$

The same reasoning can be used without memorizing the ratio formula:

1. Write $\mu = m/L$ and $v = \sqrt{F_T/\mu}$.
2. State what changes and what stays fixed.
3. Determine the factor by which μ changes.
4. Substitute that factor into the speed equation.
5. Apply the square root last.

Independent change by a factor r	Effect on μ	Effect on v
$F_T \to rF_T$	unchanged	$v \to \sqrt{r}\,v$
$m \to rm$, fixed L	$\mu \to r\mu$	$v \to v/\sqrt{r}$
$L \to rL$, fixed m	$\mu \to \mu/r$	$v \to \sqrt{r}\,v$
$m \to rm$ and $L \to rL$	unchanged	unchanged

3 Superposition, Interference, and LIGO

When two waves occupy the same place at the same time, their displacements add:

$$\boxed{y_{\text{total}} = y_1 + y_2}. \tag{6}$$

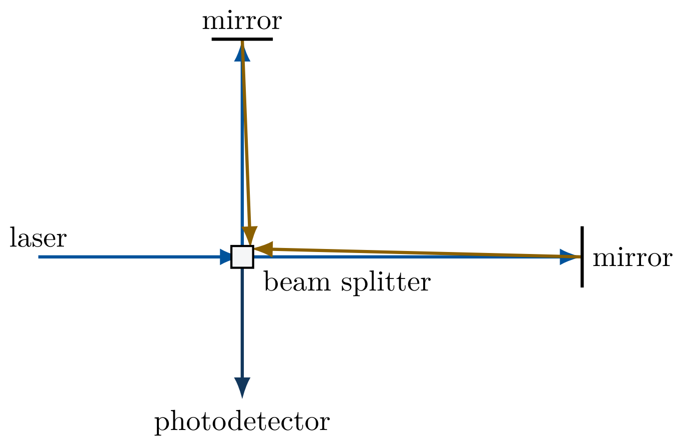
This is the **principle of superposition**.

- **Constructive interference:** crest meets crest or trough meets trough, producing a larger displacement.

- **Destructive interference:** crest meets trough, producing a smaller displacement or complete cancellation.

The **relative phase** describes how the two repeating patterns line up. Waves that return crest with crest are in phase; waves that return crest with trough are out of phase. At this level, no angle calculation is needed.

3.1 How the interferometer produces a signal



1. A laser beam is split into two beams that travel along perpendicular arms.
2. Mirrors reflect the beams back toward the beam splitter.
3. The returning beams recombine and interfere.
4. A gravitational wave changes the two arm lengths differently, so the beams travel slightly different distances.
5. The resulting change in relative phase changes the light intensity at the photodetector.

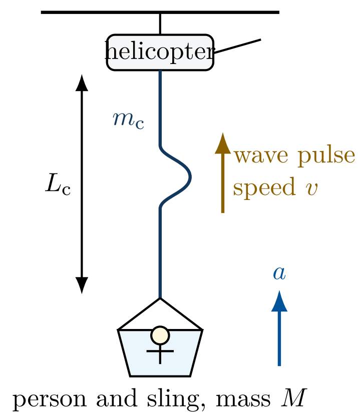
4 From a Cable Pulse to the Load's Acceleration

Suppose a helicopter lifts a person and sling with a cable of length $L_{\rm c}$ and mass $m_{\rm c}$. A wave pulse travels the full length of the cable in a measured time Δt.

The pulse does not directly tell us the acceleration of the load. Instead, it reveals the cable tension. Once the tension is known, Newton's second law gives the acceleration of the person and sling.

The calculation follows the chain

$$(L_{\rm c}, \Delta t) \longrightarrow v \longrightarrow F_T \longrightarrow a.$$



1. Find the speed of the wave pulse

The pulse travels a distance equal to the cable length:

$$\Delta x = L_c.$$

Therefore,

$$\boxed{v = \frac{\Delta x}{\Delta t} = \frac{L_c}{\Delta t}}. \tag{7}$$

2. Find the cable's linear mass density

Wave speed depends on mass per unit length, not on the cable's total mass alone. The cable's linear mass density is

$$\boxed{\mu = \frac{m_c}{L_c}}. \tag{8}$$

Only the cable's mass m_c belongs in this equation.

3. Use the wave speed to determine the tension

For a transverse wave on a cable,

$$v = \sqrt{\frac{F_T}{\mu}}.$$

Square both sides and solve for the tension:

$$v^2 = \frac{F_T}{\mu},$$
$$\mu v^2 = F_T.$$

Thus,

$$\boxed{F_T = \mu v^2}. \tag{9}$$

The units confirm that this is a force:

$$\left(\frac{\text{kg}}{\text{m}}\right)\left(\frac{\text{m}^2}{\text{s}^2}\right) = \frac{\text{kg m}}{\text{s}^2} = \text{N}.$$

In this simplified model, the cable tension is treated as uniform. The tension found from the wave speed is therefore also the upward force exerted by the cable on the load.

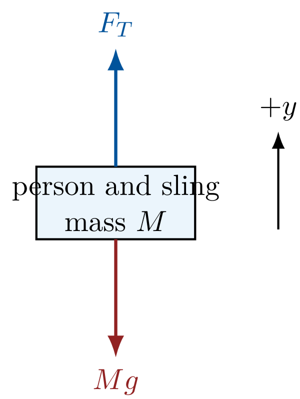
4. Use Newton's second law to find the acceleration

Now switch from analyzing the cable to analyzing the object being accelerated. Choose the person and sling together as the system. If their individual masses are m_p and m_s, then

$$\boxed{M = m_\mathrm{p} + m_\mathrm{s}}. \tag{10}$$

The external vertical forces on this system are

- the upward cable tension F_T;
- the downward weight Mg.



Choose upward as the positive direction. Newton's second law gives

$$\sum F_y = Ma,$$
$$F_T - Mg = Ma.$$

Solving for the acceleration,

$$\boxed{a = \frac{F_T - Mg}{M} = \frac{F_T}{M} - g}. \tag{11}$$

The sign of a provides a physical check:

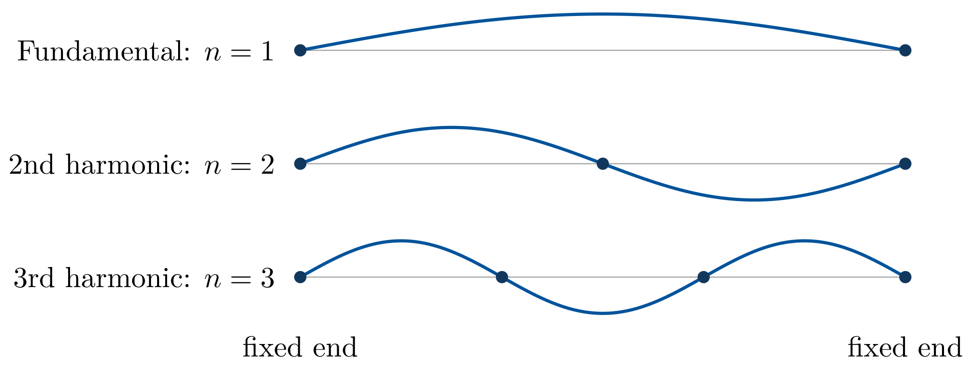
$$\begin{aligned} F_T > Mg \quad &\Longrightarrow \quad a > 0 \quad \text{(upward acceleration)},\\ F_T = Mg \quad &\Longrightarrow \quad a = 0,\\ F_T < Mg \quad &\Longrightarrow \quad a < 0 \quad \text{(downward acceleration)}. \end{aligned}$$

5 Standing Waves and Harmonics

A **standing wave** forms when waves of the same frequency travel in opposite directions and repeatedly interfere.

- A **node** is a point that never moves.
- An **antinode** is a point with maximum oscillation.

A string fixed at both ends must have a node at each end. Only certain patterns fit this boundary condition.



For the nth mode, the string length contains n half-wavelengths:

$$L = n\frac{\lambda_n}{2}.$$

Therefore,

$$\boxed{\lambda_n = \frac{2L}{n}}. \tag{12}$$

Using $v = \lambda f$,

$$\boxed{f_n = \frac{nv}{2L} = nf_1}. \tag{13}$$

Thus, the second and third harmonics have frequencies $2f_1$ and $3f_1$.

For a string, substitute $v = \sqrt{F_T/\mu}$ into $f_1 = v/(2L)$:

$$\boxed{f_1 = \frac{1}{2L}\sqrt{\frac{F_T}{\mu}}}. \tag{14}$$

5.1 Comparing the tensions of two strings

For strings A and C,

$$f_A = \frac{1}{2L_A}\sqrt{\frac{F_{T,A}}{\mu_A}}, \qquad f_C = \frac{1}{2L_C}\sqrt{\frac{F_{T,C}}{\mu_C}}.$$

If the strings have the same length and linear density, those factors cancel when the equations are divided:

$$\frac{f_A}{f_C} = \sqrt{\frac{F_{T,A}}{F_{T,C}}}.$$

Squaring both sides gives

$$\boxed{\frac{F_{T,A}}{F_{T,C}} = \left(\frac{f_A}{f_C}\right)^2}. \tag{15}$$

6 One-Page Reference

Core equations

$$
\begin{aligned}
\text{basic waves:}\quad & f = \frac{1}{T_\mathrm{p}}, & v &= \lambda f,\\
\text{string waves:}\quad & \mu = \frac{m}{L}, & v &= \sqrt{\frac{F_T}{\mu}},\\
\text{pulse on a cable:}\quad & v = \frac{L_\mathrm{c}}{\Delta t}, & F_T &= \mu v^2,\\
\text{vertical load:}\quad & F_T - Mg = Ma, & a &= \frac{F_T}{M} - g,\\
\text{fixed-end string:}\quad & \lambda_n = \frac{2L}{n}, & f_n &= \frac{nv}{2L} = nf_1,\\
\text{fundamental:}\quad & f_1 = \frac{1}{2L}\sqrt{\frac{F_T}{\mu}}.
\end{aligned}
$$

For strings with equal L and μ,

$$\boxed{\frac{F_{T,A}}{F_{T,C}} = \left(\frac{f_A}{f_C}\right)^2}.$$

Conceptual chains

- Transverse: oscillation $\perp$ wave travel.
- Longitudinal: oscillation $\parallel$ wave travel.
- LIGO: path-length difference $\rightarrow$ relative-phase change $\rightarrow$ interference change $\rightarrow$ intensity change.
- Cable problem: time $\rightarrow$ speed $\rightarrow$ tension $\rightarrow$ acceleration.
- Harmonics: $f_2 = 2f_1$, $f_3 = 3f_1$, and in general $f_n = nf_1$.